

Probability and Distributions

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice working with probability, probability distributions, and the normal distribution in R. You will work with both simulated data and spatial yield data to estimate probabilities, visualize distributions, and connect summary statistics to probabilistic statements. Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

Question 1

1 point

Load the `peanut_uniformity` dataset and assign it to an object called `peanut`.

note: this might be a good place to include a code chunk loading the `fastGraph` library

```
peanut <- read.csv('./data/peanut_uniformity.csv')
```

Question 2

1 point

Inspect the dataset using the `head()` function.

```
head(peanut)
```

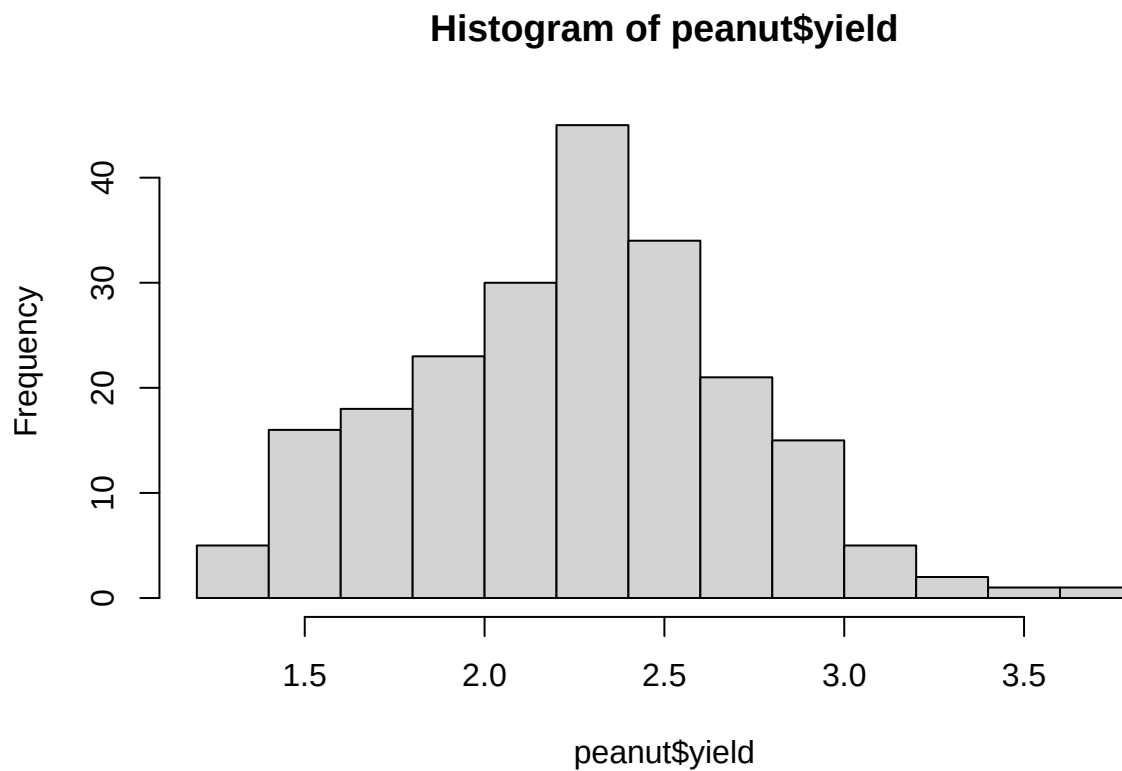
```
##   row col yield block
## 1   1   1  1.86    B1
## 2   2   1  2.32    B1
## 3   3   1  2.55    B1
## 4   4   1  1.63    B1
## 5   5   1  2.18    B1
## 6   6   1  2.55    B1
```

Question 3

1 point

Create a histogram of yield using the `hist()` function.

```
hist(peanut$yield)
```



Question 4

2 points

Calculate the mean yield and assign it to the variable `peanut_mu`. Calculate the standard deviation of yield and assign it to the variable `peanut_sigma`.

```
peanut_mu <- mean(peanut$y)
peanut_sd <- sd(peanut$y)
```

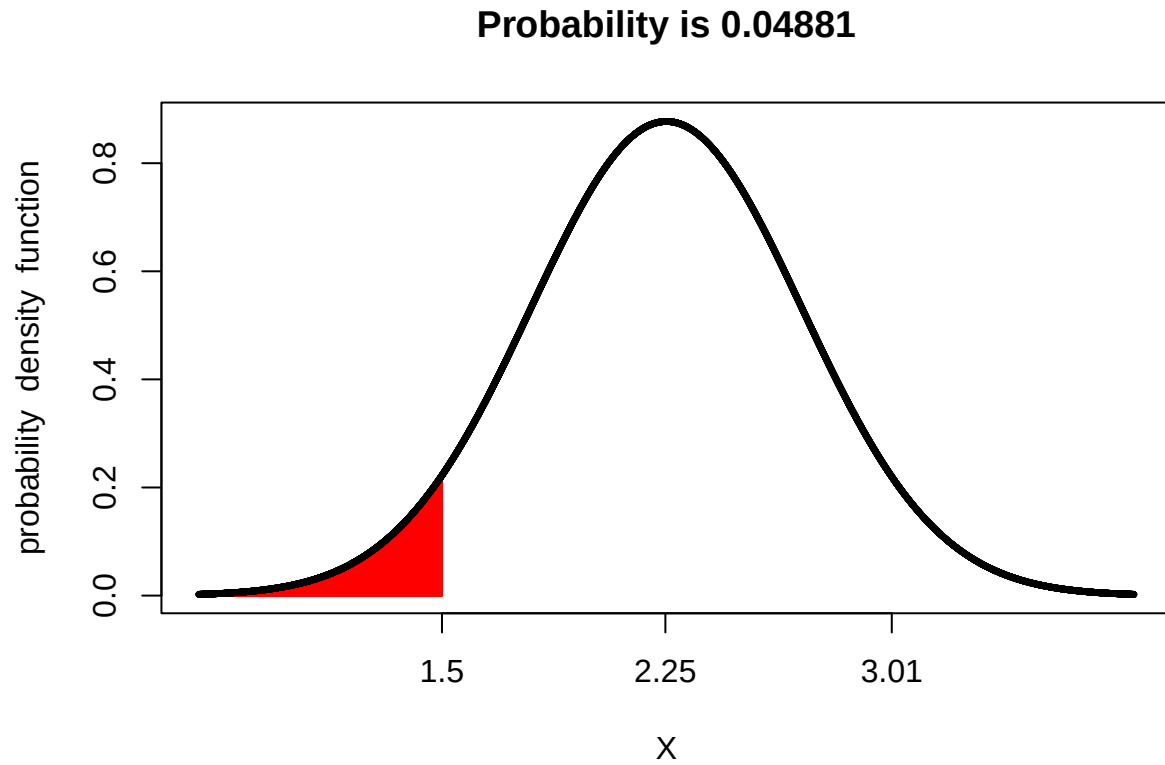
Question 5

2 points

Plot the normal distribution for the peanut yield data and calculate the probability of observing a yield less than 1.5.

```
library(fastGraph)
shadeDist(xshade = 1.5,
          ddist = 'dnorm',
```

```
parm1 = peanut_mu,  
parm2 = peanut_sd)
```

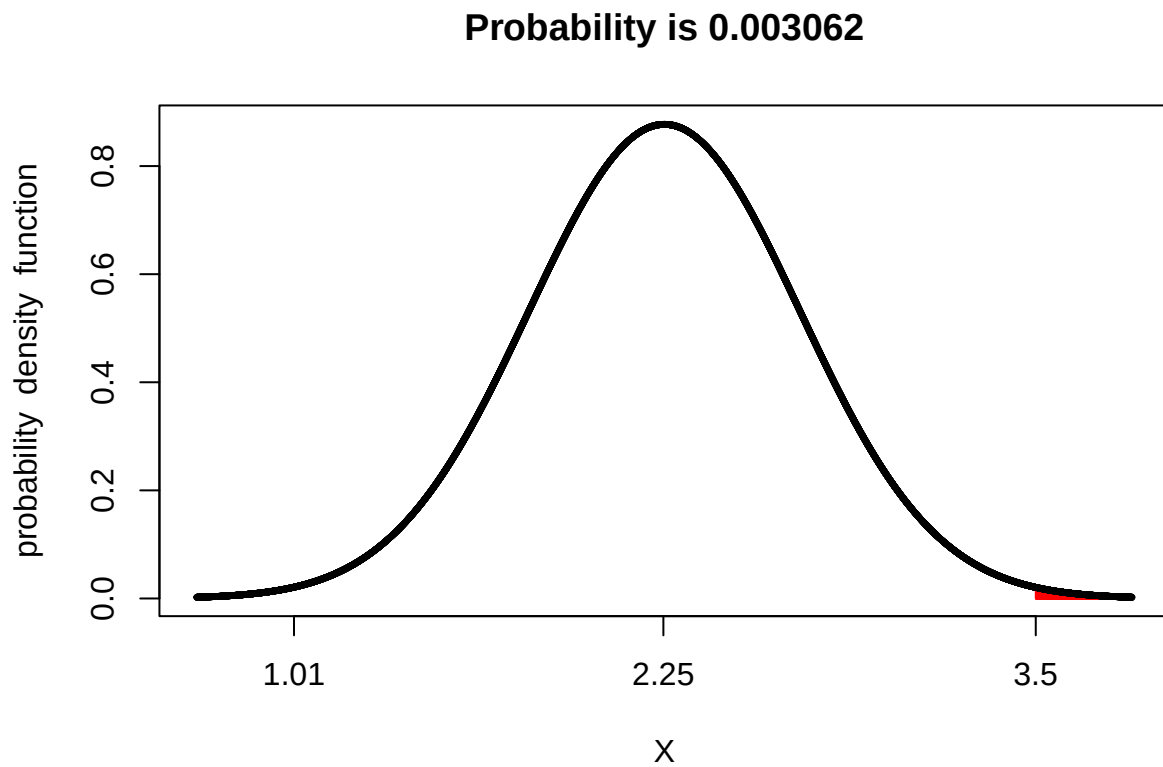


Question 6

1 point

Calculate the probability of observing a yield greater than 3.5.

```
shadeDist(xshade = 3.5,  
          ddlist = 'dnorm',  
          parm1 = peanut_mu,  
          parm2 = peanut_sd,  
          lower.tail = FALSE)
```

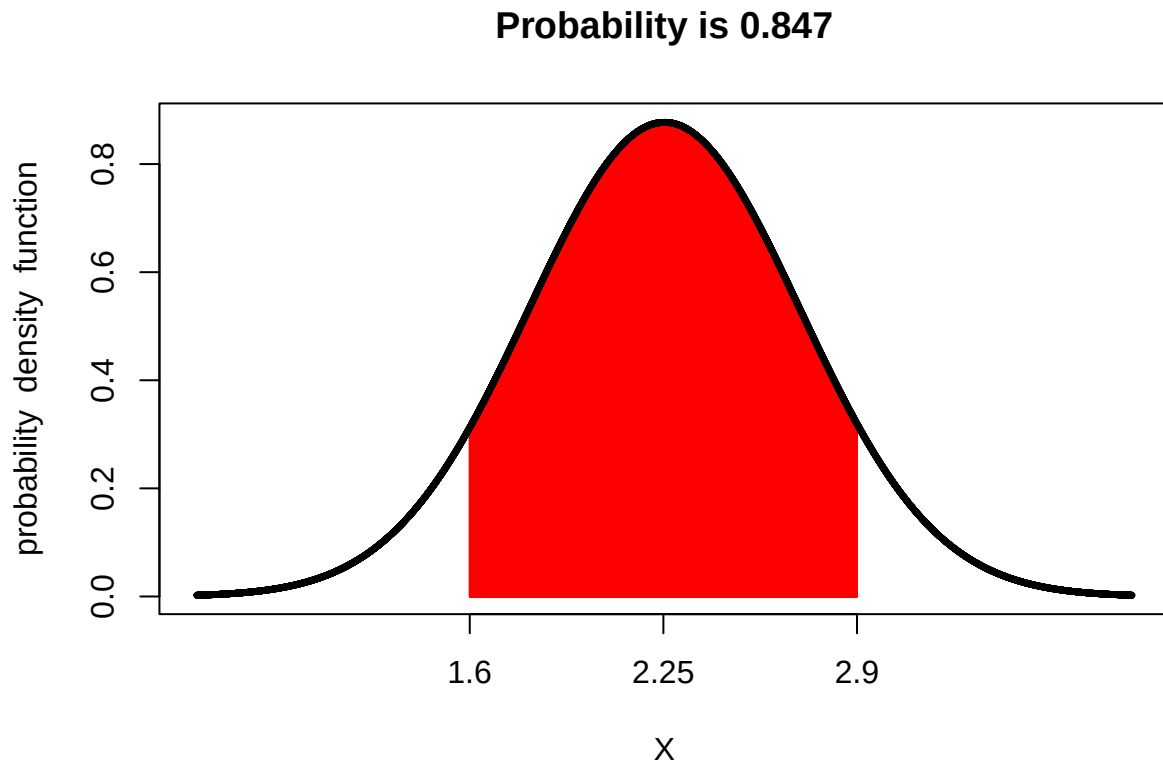


Question 7

1 point

Calculate the probability of observing a yield greater than or equal to 1.6 and less than 2.9.

```
shadeDist(xshade = c(1.6, 2.9),  
          ddist = 'dnorm',  
          parm1 = peanut_mu,  
          parm2 = peanut_sd,  
          lower.tail = FALSE)
```



Question 8

1 point

Load the shapefile "data/merriweather_yield_map/merriweather_yield_map.shp" using `st_read()` and assign it to an object called `soybean`.

note: Hear that? It sounds like there's someone saying "Load the sf package!"

```
library(sf)
```

```
## Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
soybean <- st_read('./data/merriweather_yield_map/merriweather_yield_map.shp')
```

```
## Reading layer 'merriweather_yield_map' from data source
##   '/home/caio/Dropbox/classes/2026_data-science-for-agricultural-professionals/modules/unit-02/assign
##   using driver 'ESRI Shapefile'
## Simple feature collection with 6061 features and 12 fields
## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: -93.15474 ymin: 41.66619 xmax: -93.15026 ymax: 41.66945
## Geodetic CRS:   WGS 84
```

Question 9

1 point

Inspect the `soybean` data using the `head()` function.

```
head(soybean)
```

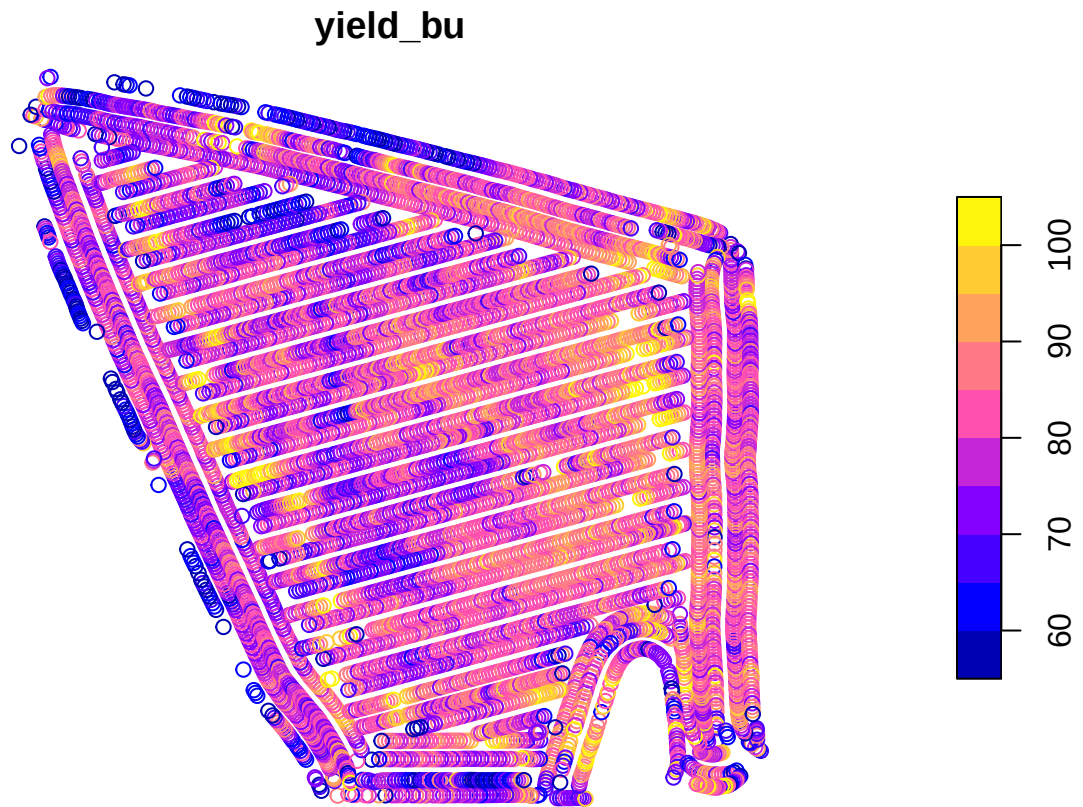
```
## Simple feature collection with 6 features and 12 fields
## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: -93.15033 ymin: 41.66641 xmax: -93.15026 ymax: 41.66644
## Geodetic CRS:   WGS 84
##   DISTANCE SWATHWIDTH VRYIELDVOL Crop  WetMass Moisture      Time
## 1 0.9202733         5    57.38461  174 3443.652     0.00 9/19/2016 4:45:46 PM
## 2 2.6919269         5    55.88097  174 3353.411     0.00 9/19/2016 4:45:48 PM
## 3 2.6263101         5    80.83788  174 4851.075     0.00 9/19/2016 4:45:49 PM
## 4 2.7575437         5    71.76773  174 4306.777     6.22 9/19/2016 4:45:51 PM
## 5 2.3966513         5    91.03274  174 5462.851    12.22 9/19/2016 4:45:54 PM
## 6 3.1840529         5    65.59037  174 3951.056    13.33 9/19/2016 4:45:55 PM
##   Heading VARIETY Elevation      IsoTime yield_bu
## 1 300.1584   23A42   786.8470 2016-09-19T16:45:46.001Z 65.97034
## 2 303.6084   23A42   786.6140 2016-09-19T16:45:48.004Z 64.24158
## 3 304.3084   23A42   786.1416 2016-09-19T16:45:49.007Z 92.93246
## 4 306.2084   23A42   785.7381 2016-09-19T16:45:51.002Z 77.37348
## 5 309.2284   23A42   785.5937 2016-09-19T16:45:54.002Z 91.86380
## 6 309.7584   23A42   785.7512 2016-09-19T16:45:55.005Z 65.60115
##           geometry
## 1 POINT (-93.15026 41.66641)
## 2 POINT (-93.15028 41.66641)
## 3 POINT (-93.15028 41.66642)
## 4 POINT (-93.1503 41.66642)
## 5 POINT (-93.15032 41.66644)
## 6 POINT (-93.15033 41.66644)
```

Question 10

2 points

Plot a yield map using the "yield_bu" column of the `soybean` dataset and create a histogram of yield.

```
plot(soybean['yield_bu'])
```



Question 11

2 points

Calculate the mean yield and assign it to the variable `soybean_mu`. Calculate the standard deviation of yield and assign it to the variable `soybean_sigma`.

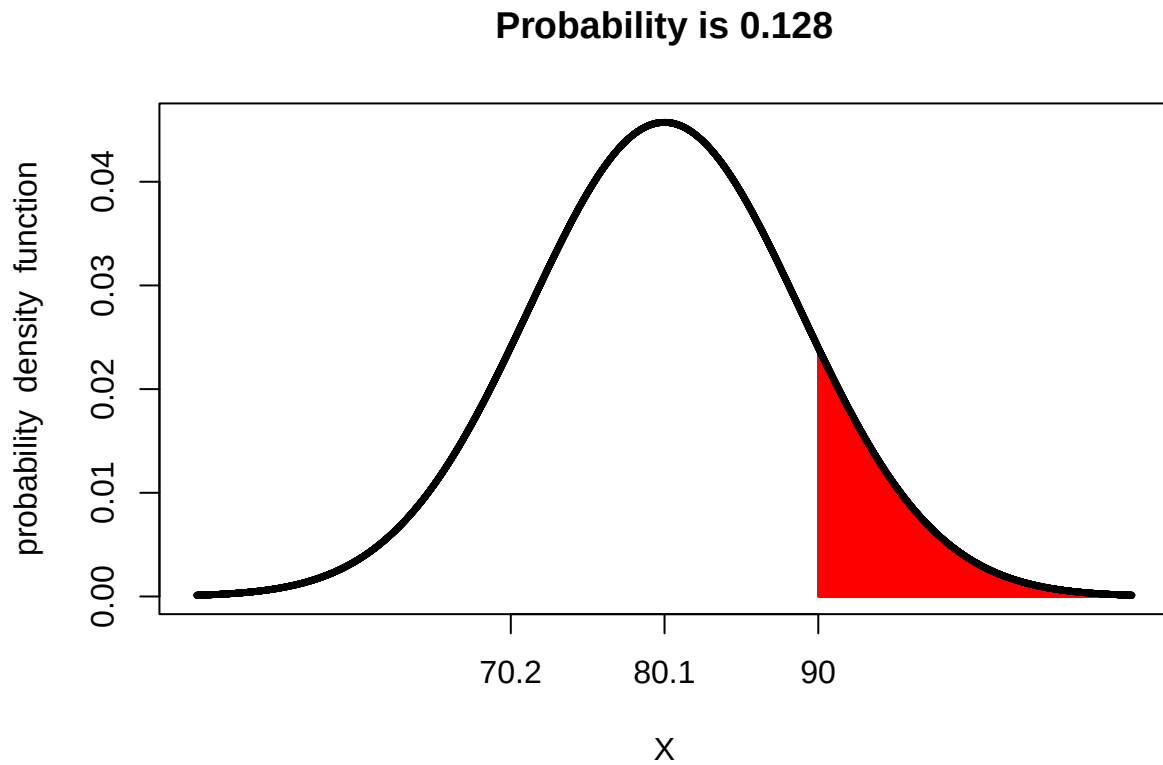
```
soybean_mu <- mean(soybean$yield_bu)
soybean_sigma <- sd(soybean$yield_bu)
```

Question 12

1 point

Using the normal distribution, calculate the probability of observing a yield greater than or equal to 90 bu/ac.

```
shadeDist(xshade = 90,
          ddist = 'dnorm',
          parm1 = soybean_mu,
          parm2 = soybean_sigma,
          lower.tail = FALSE)
```



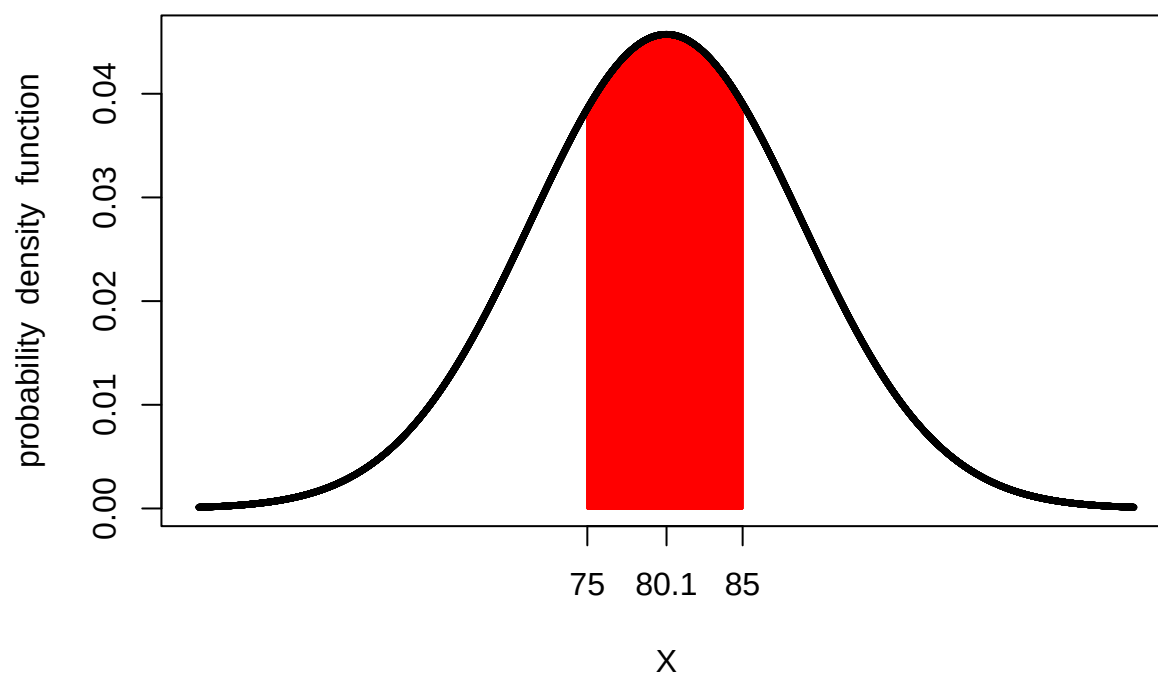
Question 13

1 point

Using the normal distribution, calculate the probability of observing a yield between 75 and 85 bu/ac.

```
shadeDist(xshade = c(75, 85),  
          ddist = 'dnorm',  
          parm1 = soybean_mu,  
          parm2 = soybean_sigma,  
          lower.tail = FALSE)
```

Probability is 0.4334



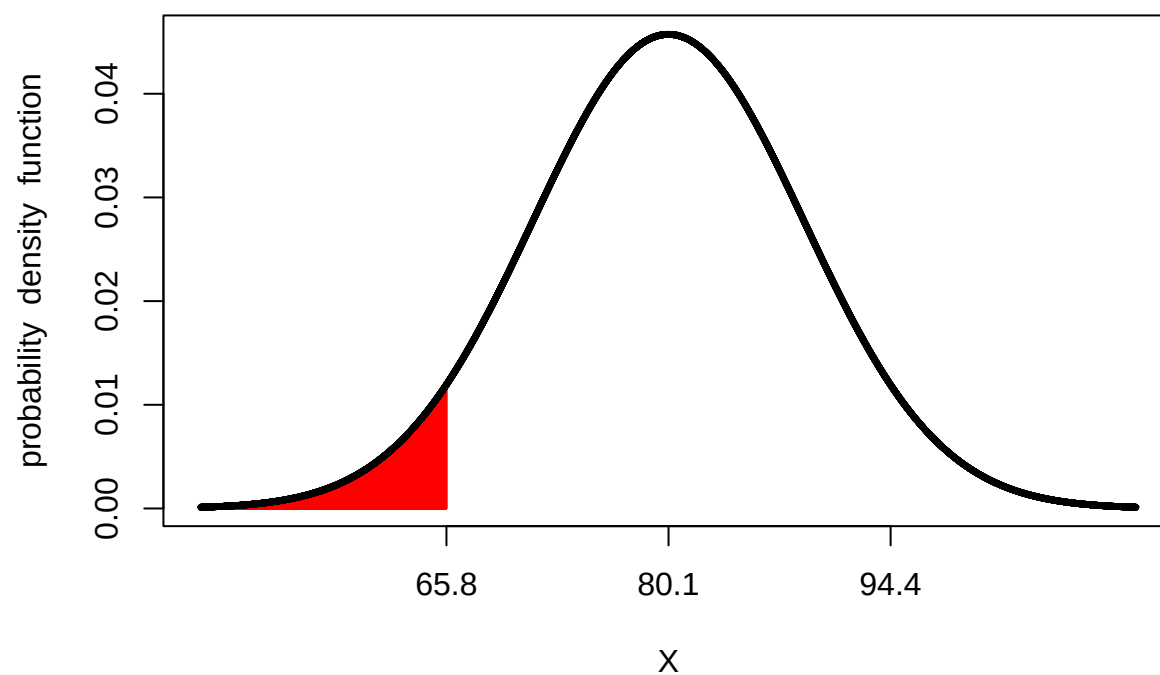
Question 14

1 point

If the probability of observing a yield less than 65.8 bu/ac is 5%, about how many observations out of 100 would you expect to be less than or equal to 65.8 bu/ac?

```
shadeDist(xshade = 65.8,  
          ddist = 'dnorm',  
          parm1 = soybean_mu,  
          parm2 = soybean_sigma,  
          lower.tail = TRUE)
```

Probability is 0.0507



Write your brief interpretation here:

I would expect 5 observations out of 100 to be under 65.8 bu/ac.